Name: _____

Date: _____

Mr. Rawson • Y9 MYP Physics

2.5 – Resistance

Unit 2: Electricity & Energy

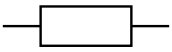
The question: *you've measured current and voltage — can you use the two of them together to say exactly how much a component resists the flow?* Today, one formula links all three.

Learning intentions

All students will state the formula linking voltage, current and resistance, and use it to find **resistance**.
Most students will rearrange it to find **voltage** and **current** too.
Some students will calculate the **total** resistance of two resistors in series.

Part 1 — Resistance, and finding it

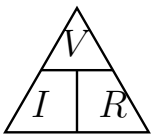
A **resistor**'s job is to resist current, on purpose. Resistance is measured in **ohms** (Ω).



Voltage, current and resistance are linked by one formula:

$$V = I \times R$$

The triangle: cover the letter you want, and the other two tell you what to do with it.



Cover R: you're left with V over I — so $R = \dfrac{V}{I}$.

1) **Calculate** the resistance of a bulb, if the voltage across it is **6 V** and the current through it is **0.5 A**.

Model answer: $R = \dfrac{V}{I} = \dfrac{6}{0.5} = 12\,\Omega$.

Build it

Build a circuit with one resistor, an ammeter and a voltmeter. Record both readings, then calculate the resistance.

$V =$ _____ V $I =$ _____ A $R = \dfrac{V}{I} =$ _____ Ω

Check: compare your answer with the resistor's printed or stated value — they should be close.

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Part 2 — Rearranging the same formula

Cover V on the triangle: you're left with I and R side by side — so $V = I \times R$, the formula you already know.

2) **Calculate** the voltage across a **20 Ω** resistor, if the current through it is **0.3 A**.

Model answer: $V = I \times R = 0.3 \times 20 = 6\,V$.

Cover I: you're left with V over R — so $I = \dfrac{V}{R}$.

3) **Calculate** the current through a **15 Ω** resistor, if the voltage across it is **3 V**.

Model answer: $I = \dfrac{V}{R} = \dfrac{3}{15} = 0.2\,A$.

Mixed practice

Fill in the missing value in each row. Show the version of the triangle you used.

V	I	R
12 V	2 A	6 Ω
9 V	0.5 A	18 Ω
4 V	0.8 A	5 Ω

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Some students — two resistors in series

In series, each resistor adds its own resistance to the total — there's simply more to push through:

$$R_{\text{total}} = R_1 + R_2$$

4) **Calculate** the total resistance of a **12 Ω** resistor and a **8 Ω** resistor, connected in series. Then calculate the current that would flow if a **10 V** cell were connected across them.

Model answer: $R_{total} = 12 + 8 = 20\,\Omega$. *Then* $I = \frac{V}{R} = \frac{10}{20} = 0.5\,A$.

Extension — what changes a wire's resistance

A wire's resistance depends on three things: a **longer** wire has more resistance, a **thicker** wire has less resistance, and some **materials** (like copper) resist current far less than others (like nichrome).

5) **State** which single change — longer, shorter, thicker or thinner — would you make to a wire to *increase* its resistance the most, keeping the material the same?

Model answer: *Make it longer and thinner at the same time increases resistance the most; if only one change is allowed, making it longer (or thinner) both increase resistance — either is an acceptable single answer.*

Part 3 — On your own

A torch bulb needs **0.3 A** at **3 V** to light correctly. The torch's battery pack supplies **4.5 V** — too much — so a **variable resistor** is added in series to use up the spare voltage.

6) **Calculate** the resistance the variable resistor must be set to, so the bulb gets exactly 3 V and 0.3 A flows through the whole circuit.

Model answer: *The variable resistor must carry the leftover voltage:* $4.5 - 3 = 1.5\,V$. *The current through it is the same 0.3 A (series circuit, current rule from L3). So* $R = \frac{V}{I} = \frac{1.5}{0.3} = 5\,\Omega$.

HOMEWORK

Using the triangle, calculate the missing value:

1. $V = 10\,\text{V}$, $R = 4\,\Omega$. Find I.
2. $I = 1.5\,\text{A}$, $R = 6\,\Omega$. Find V.
3. $V = 24\,\text{V}$, $I = 3\,\text{A}$. Find R.

Show all working — the triangle version you used for each.